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Effects of Biochar Application on Tomato Yield and Fruit Quality: A Meta-Analysis

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Abstract: Applying biochar to tomato cultivation presents a beneficial strategy that can enhance both yield and fruit quality, crucial for sustainable agricultural practices. However, a review of the existing literature on the effects of biochar indicates a significant variability in outcomes, suggesting the need for a more nuanced understanding of biochar application in relation to soil and biochar conditions. This study conducts a meta-analysis on the literature published before March 2024 to investigate the impacts of biochar properties, agricultural practices, and soil properties on the yield and fruit quality of tomato. The results indicated that biochar application significantly increased tomato yield by 29.55%, total soluble solids (TSS) by 4.28%, and vitamin C (VC) by 6.77% compared to control treatments without biochar, especially at higher application rates. However, the benefits may wane over time due to biochar aging in the soil, requiring periodic replenishment. The type of biochar and pyrolysis temperature, particularly wood and straw biochar pyrolyzed at 401–500 °C, were found to be most effective for boosting yield and quality. Additionally, initial soil properties, including soil organic matter, pH, and nutrient levels, interact with biochar to influence outcomes, with biochar being particularly beneficial for soils with a high bulk density and low soil organic matter (SOM) or nutrient deficiencies. This study underscores the potential of biochar as a multifaceted strategy in tomato cultivation, enhancing not only yield but also the nutritional value of the fruit, while simultaneously improving soil health.

Keywords: biochar; meta-analysis; tomato; yield; fruit quality

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Citation: Lei, Y.; Xu, L.; Wang, M.; Sun, S.; Yang, Y.; Xu, C. Effects of Biochar Application on Tomato Yield and Fruit Quality: A Meta-Analysis. *Sustainability* **2024**, *16*, 6397. https://doi.org/10.3390/su16156397

Academic Editor: Kęstutis Venslauskas

Received: 18 June 2024
Revised: 19 July 2024
Accepted: 22 July 2024
Published: 26 July 2024

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1. Introduction

Tomato is the second largest vegetable crop in the human diet worldwide, with tremendous economic and nutritional importance [1]. Due to its culinary value and high nutrient content, it occupies a vital position among vegetables [2]. In 2022, the global production of tomato was 186.1 million tons (t), and the global tomato area harvested was 4.7 million hectares [3]. Tomatoes serve as the principal source of high-quality antioxidants like vitamin C (VC) in our diets, which are linked to a range of health advantages, including a reduced risk of heart disease and cancer [4,5]. However, irrational fertilizer application and cultivation practices have led to an increased vulnerability to pests and diseases in both greenhouse and field soils, which have seriously impacted the growth and development of tomatoes, leading to a decrease in both the quantity and quality of the fruit [6]. Therefore, the future must employ innovative approaches and technologies to attain substantial agricultural production while mitigating adverse effects on the soil.

Among various emerging strategies, biochar stands out as a viable option for sustainable agriculture because, so far, it has been shown that its incorporation into soil substantially alters the physicochemical properties together with the plant characteristics [7]. Biochar is a kind of organic substance rich in carbon, resulting from the pyrolysis of biomass such as wood, plant residues, and agricultural waste under high temperatures

in a hypoxic environment [8]. The pyrolysis process, which is a form of thermochemical treatment, transforms organic biomass into biochar, endowing it with a large surface area and a distinctive capacity to endure in various soils with a biological degradation rate that can span from several decades to centuries [9]. Concerning the impact of biochar on plant growth, biochar serves as an effective soil amendment, capable of enhancing soil fertility, promoting plant growth through nutrient provision, and ensuring the retention of these nutrients within the soil [10,11]. Moreover, there is evidence that high doses of biochar may cause soil salinity issues, particularly in alkaline soils by increasing the pH, leading to nutrient precipitation. Additionally, certain studies have indicated that the application of biochar can improve the biological properties and bolster the abundance and vitality of the soil's microbial community [12,13]. Simultaneously, the application of biochar on tomato increases the specific root length and favors fine root proliferation, which enhances the roots' ability to adsorb nutrients and improves nitrogen uptake [14,15]. The practical implications of these benefits are profound for farmers and agricultural practitioners, as biochar's ability to improve soil fertility and water retention can lead to more resilient crop production systems that require less external input, such as synthetic fertilizers and irrigation water. Recycling pyrolyzed plant residues to the field is a beneficial approach that maximizes the utilization of agricultural by-products and enhances the agricultural ecosystem. Consequently, the incorporation of biochar on tomato can be seen as a multi-win strategy.

Many studies have explored the impact of biochar on tomato cultivation. However, there remains considerable fluctuation and unpredictability regarding its effects on tomato yield and fruit quality. While certain studies have documented a significant increase in tomato yields through the conversion of agricultural residues into biochar [16–18], other research has suggested that biochar application may not enhance tomato yields and could even have an adverse effect [19–21]. Regarding sugar/acid, several research studies have found it was decreased by biochar application [22]; in contrast, some studies have believed biochar has a positive effect [23]. Many studies have indicated that biochar application can improve the total soluble solids (TSS) and VC content [24–26], yet other research has also highlighted adverse effects on these parameters [27]. The heterogeneity observed in these outcomes can be attributed to a multitude of factors, including soil properties, experimental settings, and biochar characteristics. The composition and characteristics of biochar were influenced by the variety of feedstock sources and pyrolysis temperatures [28]. Biochar which was produced from woody and crop residue biomass exhibited a higher carbon content when compared to biochar sourced from manure [29], and in general, the C/N ratio, pH, and ash content of biochar were observed to increase as pyrolysis temperatures rose [9,30]. Biochar, being a carbon-rich substrate with an elevated C/N ratio, triggered soil microorganisms to decompose the native soil organic matter (SOM) in order to assimilate nitrogen through a priming effect when it was applied to the soil [31]. Biochar ash is rich in nutrients that foster plant growth, and its high ash content and pH levels have the potential to alter the soil's pH balance. The original soil pH also plays an important role in determining how biochar impacts the behavior of soil microbes and the alteration of certain nutrients. Furthermore, Hussain et al. [32] have reported that the application of biochar has been associated with increased yields, particularly in severely depleted and nutrient-deficient soils. Based on the literature review provided, the aforementioned parameters have not undergone a systematic examination. Therefore, the purpose of this study is to conduct a comprehensive analysis of biochar application on tomato, focusing on the impact of influential factors on tomato yield and fruit quality.

A meta-analysis represents a valuable method for amalgamating data from various individual studies, thereby offering a comprehensive measure of the overall treatment effects [33]. By integrating numerous data sources into a unified dataset through the use of existing observational data in experimental studies, a meta-analysis is capable of minimizing experimental errors and delivering a more accurate estimation of the treatment effect size [34,35]. It is also regarded as an effective strategy to address the issue of variability observed among various individual studies [6,36]. To address the variability in

outcomes, our meta-analysis is framed around the hypothesis that biochar application significantly improves tomato yield and fruit quality. We specifically aim to answer the following: (1) How does biochar application impact tomato yield and nutritional quality? and (2) What biochar characteristics and environmental conditions optimize these benefits? The results of this meta-analysis can provide a theoretical basis for the selection of biochar for different purposes in tomato cultivation and sustainable green agriculture.

2. Materials and Methods

2.1. Data Collection

A comprehensive survey of the literature was conducted, primarily utilizing databases including ISI Web of Science, Science Direct, and the China Knowledge Resource Integrated Database. To collect relevant studies, the search terms "biochar" and "tomato yield or fruit quality" were utilized for articles published before March 2024. To mitigate publication bias, 46 articles were found and screened, according to the inclusion criteria for the meta-analysis (refer to Supplementary Material) as follows: (1) At least tomato yield or one of the collected quality indicators (sugar/acid, TSS, VC) was reported; (2) Each selected study was executed in soil environments, not in horticultural substrates; (3) The experiment should encompass at least three replicates for each treatment, adhering to a randomized experimental layout; (4) The study should incorporate both control and treatment groups, with the treatment involving biochar application and the control group featuring identical experimental conditions but without biochar.

The data shown in graphs or figures were extracted using the GetData Graph Digitizer version 2.22. There were two calculation methods for missing SD: (1) if standard error (SE) was given, SD could be converted using SE by the following formula:

$$SD = SE \times \sqrt{n} \quad (1)$$

(2) otherwise, SD was calculated by R version 4.31 package metagear 0.7 [37], using the method of Bracken to input the missing SD [38].

To elucidate the factors that affect the effects of biochar on tomato yield and fruit quality, we compiled data on the experimental conditions, environmental factors, and biochar characteristics from each study. Subsequent subgroup analyses were conducted, including the following:

1. experimental site (greenhouse and field);
2. biochar application rate ($\leq$5, 5~20, 20~50, 50~80, >80 t·hm^{-2});
3. experimental duration ($\leq$0.5, 0.5~1.0, >1 year);
4. type of biochar (wood, straw, shell residue, sludge, mixed);
5. pH of biochar ($\leq$8, 8~10, >10);
6. pyrolysis temperature of biochar ($\leq$400, 401~500, 501~600, >600 °C);
7. C/N of biochar ($\leq$20, 20~50, 50~100, 100~300, >300);
8. soil pH ($\leq$6.5, 6.5~7.5, >7.5);
9. SOM ($\leq$9.44, 9.44~16.80, >16.80 g·kg^{-1});
10. electrical conductivity (EC, $\leq$142.0, 142.0~185.0, >185.0 μS·cm^{-1});
11. soil available potassium ($\leq$101.0, 101.0~155.6, >155.6 mg·kg^{-1});
12. soil available phosphorus ($\leq$12.60, 12.64~81.26, >81.26 mg·kg^{-1});
13. soil total nitrogen ($\leq$0.96, 0.96~1.85, >1.85 g·kg^{-1});
14. soil bulk ($\leq$1.38, 1.38~1.41, >1.41 g·m^{-3})

2.2. Meta-Analysis

The effects of biochar treatment on tomato yield, sugar/acid, TSS, and VC were estimated via meta-analysis. lnRR was the natural log response ratio of the means of the biochar application treatment (X_t) to that of the no biochar application treatment (X_c) [39]:

$$\text{lnRR} = \ln\left(\frac{X_t}{X_c}\right) \quad (2)$$

The variance of the natural log response ratio (lnRR) is a critical statistical measure in a meta-analysis that quantifies the variability of the treatment effect estimates. The formula for the variance of lnRR is given as follows [39]:

$$v = \frac{SD_t^2}{n_t \times X_t^2} + \frac{SD_c^2}{n_c \times X_c^2} \quad (3)$$

where SD_t and SD_c are the standard deviations of biochar treatments and the control, respectively; and n_t and n_c are the number of replicates for biochar treatments and the control, respectively [39]. This formula accounts for the within-group variability (as indicated by the standard deviations) and the sample size of each group and allows us to accurately estimate the combined effect size across different studies while accounting for the variability inherent in each study's data.

The combined effect size over all available studies was estimated with a random effects model by R software version 4.31 package metafor 4.6. The random effects model was selected on the basis that we acknowledged the potential heterogeneity in true effect sizes across the included studies, which could be attributed to varying study conditions and environmental factors. Additionally, our aim was to extend the applicability of our findings to a broader context beyond the specific studies examined [39]. The random effects model was estimated with the DerSimonian–Laird estimator [40]. Each study's weight was determined by the reciprocal of its sampling error variance, a method known as inverse-variance weighting. The effect size and 95% confidence interval (CI) were calculated for each set of data. When the 95% confidence interval (CI) of any effect size overlaps with 0, it is considered that the treatment has no significant statistical meaning for the outcome indicator. Otherwise, it is considered that the treatment has a significant statistical meaning for the outcome indicator. The RR, standard error, and 95% CI expressed in natural log were converted to percentage change by the following equation:

$$\text{Effect size (\%)} = (\exp(\text{lnRR}) - 1) \times 100\% \quad (4)$$

2.3. Publication Bias

To determine if there was a potential publication bias concerning the target variables across studies, we utilized Egger's regression test [41] for assessing the symmetry of funnel plots and also applied Rosenthal's fail-safe number (Nfs) as additional criteria [42].

If the data pass Egger's regression test ($p > 0.05$), the funnel plot was symmetrical, and there was no publication bias. Nfs tests the potential number of non-significant effect values to render the effect value of a certain study result non-significant. If the Nfs is larger than 5K + 10, where K is the number of studies, it indicates that the conclusion is reliable, and the bias has little impact.

The funnel plots of tomato sugar/acid, TSS, and VC content were symmetrical. The Nfs of tomato yield was 11,457, which was significantly higher than (5k + 10). Overall, the outcomes of testing publication bias showed that there was no publication bias; the results of this meta-analysis were steady.

3. Results

3.1. Overview of the Dataset

Supplementary Material offers an overview of the articles included in this meta-analysis, with 46 articles and 229 individual studies satisfying our selection criteria. Tomato yield, TSS, sugar/acid, and VC were all generally normally distributed but varied greatly among the studies (Figure 1). Based on the compiled research, the application of biochar resulted in a significant enhancement in tomato yield, TSS, and VC content compared to the control treatment, while sugar/acid was not significantly varied, as illustrated in Figure 2.

The application of biochar generally led to an increase in tomato yield, TSS, and VC by 29.55% (95% CI: 25.33% to 33.91%), 4.28% (95% CI: 1.68% to 6.96%), and 6.77% (95% CI: 2.80% to 10.88%), respectively.

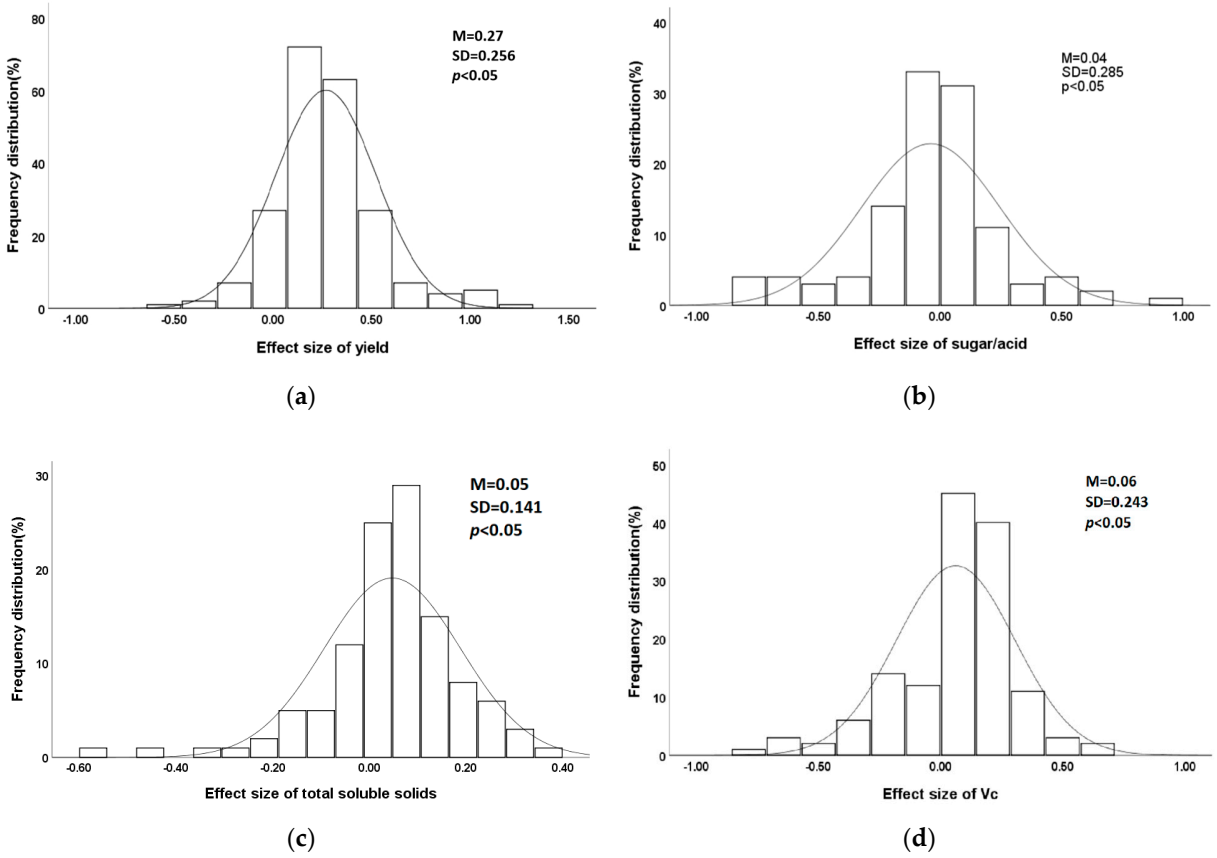


Figure 1. Frequency distributions of effect size: (**a**) effect size of yield; (**b**) effect size of sugar/acid; (**c**) effect size of TSS; (**d**) effect size of VC.

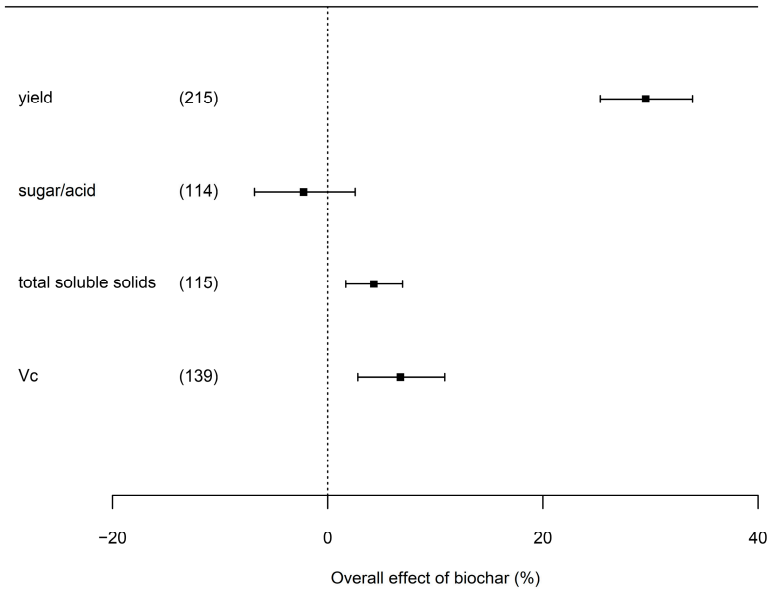


Figure 2. A forest plot shows the percentage changes (%) of biochar effects on tomato yield, sugar/acid, TSS, and VC in response to biochar application. Bars indicate 95% confidence intervals. Data in parenthesis represent the number of observations.

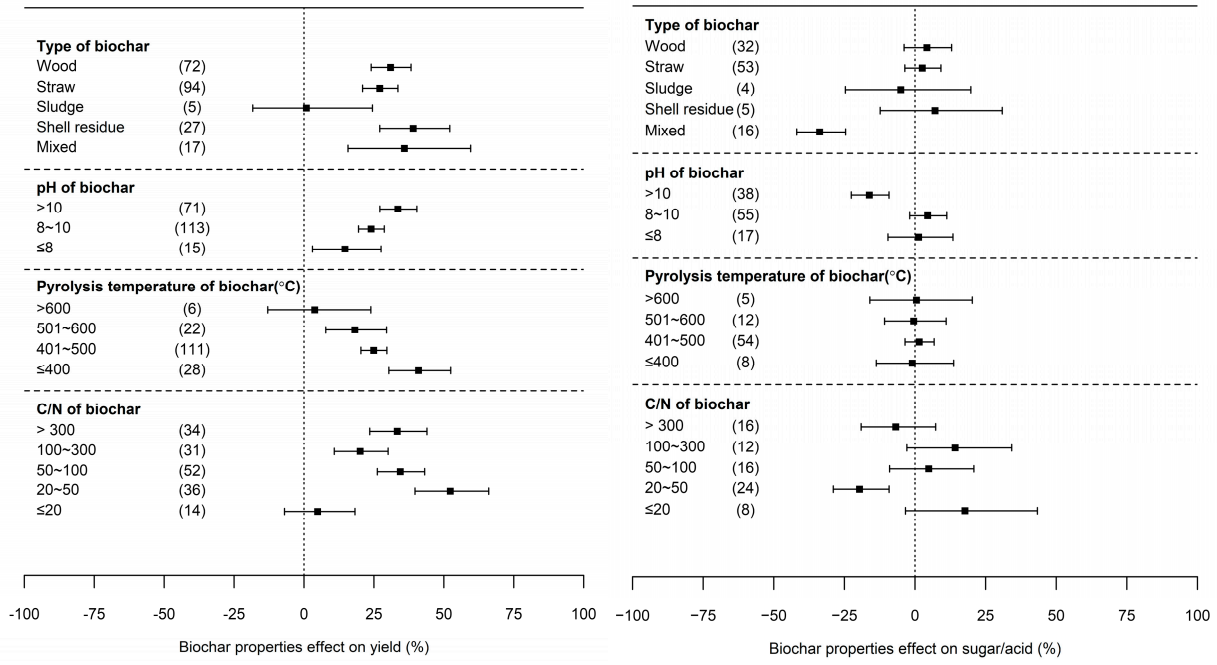
3.2. Meta-Analysis of Biochar Properties

Figure 3a shows the effects of the biochar properties on yield. The types of biochar feedstocks identified in the reviewed articles were categorized into five distinct groups: wood, straw, sludge, shell residue, and mixed. The biochar produced from wood, straw, shell residue, and mixed feedstocks improved the tomato yield significantly, while the group of sludge was not significantly varied. The tomato yield increased by 14.63% to 33.58%, depending on the pH of the biochar. The greatest promotion of yield was observed at a biochar pH above 10 (33.58%; 95% CI = 27.12%, 40.36%), followed by biochar pH 8~10 (24.01%; 95% CI = 9.50%, 28.70%), and ≤8 (14.63%; 95% CI = 3.02%, 27.55%). Upon the application of biochar, aside from biochar pyrolyzed at temperatures >600 °C, the biochar produced at pyrolysis temperatures ≤400 °C, 401~500 °C, and 501~600 °C exhibited significant positive impacts on tomato yield, resulting in an increase of 40.97% (95% CI = 30.34%, 52.47%), 24.91% (95% CI = 20.35%, 29.64%), and 18.16% (95% CI = 7.78%, 29.53%), respectively, relative to the control treatment. Regarding the C/N in biochar, the increase in yield was significant at 20~50 (52.3%; 95% CI = 39.69%, 66.05%), 50~100 (34.41%; 95% CI = 26.22%, 43.14%), 100~300 (20.06%; 95% CI = 10.82%, 30.07%), and >300 (33.34%; 95% CI = 23.51%, 43.97%), but not significant at ≤20 (4.87%).

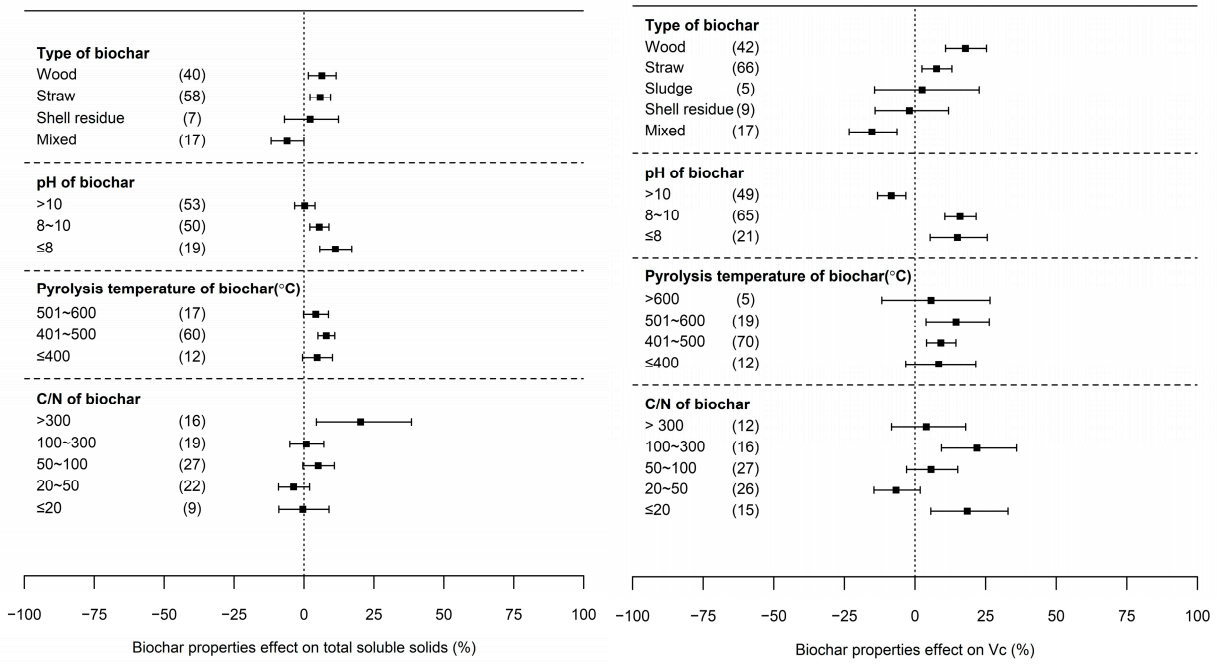
Figure 3b displays the effects of the biochar properties on the sugar–acid ratio. Among the types of biochar, biochar produced from mixed material significantly decreased the sugar–acid ratio of tomato fruits (−33.79%; 95% CI = −41.88%, −24.57%), while other types did not vary significantly. Regarding the pH of biochar, the decrease in the sugar–acid ratio was significant at >10 (−16.15%; 95% CI = −22.57%, −9.2%), but not significant in the other groups. Among the pyrolysis temperatures of biochar, all groups had no significant effects on the sugar–acid ratio. Regarding the C/N of biochar, the decrease in the sugar–acid ratio was significant at 20~50 (−19.66%; 95% CI = −28.94%, −9.18%), but not significant in other groups.

The effects of the biochar properties on the TSS of tomato fruits are given in Figure 3c. The biochar produced from wood and straw improved the TSS significantly by 6.4% (95% CI = 1.55%, 11.49%) and 5.77% (95% CI = 2.17%, 9.51%), while the TSS of the mixed group was significantly reduced by the application of biochar (−6.07%; 95% CI = −11.77%, −0.01%). When the pH of biochar was at ≤8 and 8~10, TSS was significantly increased by 11.21% (95% CI = 5.64%, 17.08%) and 5.45% (95% CI = 2.09%, 8.93%), but not significant at >10 (0.24%). After biochar was applied, the biochar produced at the pyrolysis temperature 401~500 °C improved TSS by 7.95% (95% CI = 4.97%, 11.01%), while the groups of ≤400 °C and 501~600 °C were not significant. When the C/N of biochar was >300, TSS was significantly increased by biochar application (20.24%; 95% CI = 4.43%, 38.43%), while other groups were not significantly varied.

As illustrated in Figure 3d, this meta-analysis indicates that the effects of biochar application on the VC content of tomato fruits varied, corresponding to the biochar properties. Biochar produced from wood and straw material significantly increased VC by 17.85% (95% CI = 10.82%, 25.32%) and 7.64% (95% CI = 2.46%, 13.07%). In contrast, a significant decrease was observed in the group of mixed material (−15.26%; 95% CI = −23.31%, −6.37%). VC was significant increased by the biochar with a pH ≤ 8 (15.02%; 95% CI = 5.36%, 25.58%) and 8~10 (15.97%; 95% CI = 10.57%, 21.64%), but decreased at >10 (−8.38%; 95% CI = −13.26%, −3.24%). Among the pyrolysis temperatures of biochar, the increase in VC was significant at 401~500 °C (9.17%; 95% CI = 4.09%, 14.5%) and 501~600 °C (14.56%; 95% CI = 3.91%, 26.3%), but was not significant at ≤400 °C (8.42%) and >600 °C (5.69%). When the C/N of biochar was ≤20 and 100~300, biochar application significantly increased VC by 18.48% (95% CI = 5.60%, 32.94%) and 21.95% (95% CI = 9.35%, 35.99%), respectively, while other groups were not significantly varied.



(**a**) (**b**)



(**c**) (**d**)

Figure 3. A forest plot shows the percentage changes (%) of biochar effects on (**a**) tomato yield, (**b**) sugar/acid, (**c**) TSS, and (**d**) VC in response to different biochar properties. Bars indicate 95% confidence intervals. Data in parenthesis represent the number of observations.

3.3. Meta-Analysis of Agriculture Practices

Figure 4a displays the effects of agriculture practices on tomato yield. Biochar application in the greenhouse and field groups significantly elevated the tomato yield by 30.93% (95% CI: 25.18% to 36.94%) and 32.77% (95% CI: 19.36% to 47.69%), respectively. The effects of biochar on tomato yield escalated with higher rates of biochar application. The maximum

yield was observed in groups where the biochar application exceeded 80 t·hm^{-2} (38.53%; 95% CI = 29.73%, 47.94%). With an application rate below 5 t·hm^{-2}, the increase was not significant (9.77%; 95% CI = −1.21%, 21.98%). Biochar applications in all experimental duration groups significantly increased the yield by 22.19% (95% CI: 13.84% to 31.14%), 27.59% (95% CI: 19.50% to 36.22%), and 33.72% (95% CI: 26.74% to 41.08%), respectively.

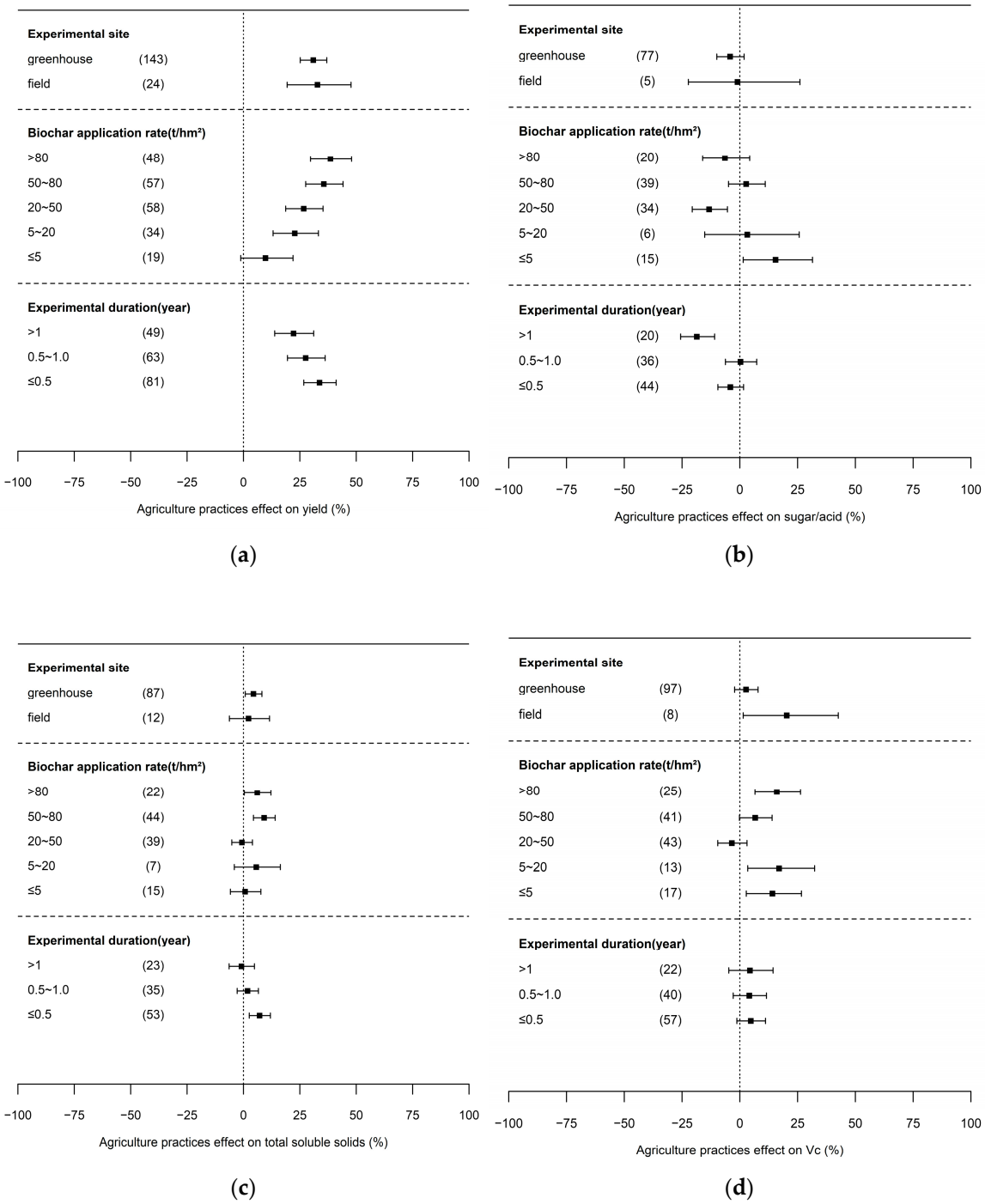


Figure 4. A forest plot shows the percentage changes (%) of biochar effects on (**a**) tomato yield, (**b**) sugar/acid, (**c**) TSS, and (**d**) VC in response to different agriculture practices. Bars indicate 95% confidence intervals. Data in parenthesis represent the number of observations.

As depicted in Figure 4b, the meta-analysis conducted indicates that the effects of biochar application on the sugar/acid of tomato fruits varied, corresponding to the agri-

cultural practices. In the groups of greenhouse and field, biochar application decreased the sugar/acid of tomato fruits but not significantly. When the biochar application rate was smaller than 5 t·hm^{-2}, sugar/acid was significantly increased by the application of biochar (15.49%; 95% CI = 1.46%, 31.45%). In contrast, a significant decrease was observed at 20~50 t·hm^{-2} (−13.31%; 95% CI = −20.59%, −5.36%). When the experimental duration was more than 1 year, the application of biochar significantly decreased the sugar/acid of tomato fruits by −18.59% (95% CI: −25.62% to −10.91%), while other groups did not vary significantly.

The effects of the agricultural practices on the TSS of tomato fruits are given in Figure 4c. Among the experimental sites, the increase in TSS was significant in the greenhouse group (4.43%; 95% CI = 0.80%, 8.18%), but not significant in the field group (2.28%). When the application rate of biochar was at 50~80 and >80 t·hm^{-2}, TSS was significantly increased by 9.14% (95% CI: 4.43% to 14.05%) and 6.12% (95% CI: 0.39% to 12.18%), respectively, while the variation in the other groups was not significant. When the experimental duration was less than 0.5 year, the application of biochar significantly increased TSS by 7.16% (95% CI: 2.57% to 11.94%), while other groups did not vary significantly.

As illustrated in Figure 4d, the conducted meta-analysis displays that the effects of biochar application on the VC content of tomato fruits varied, corresponding to the agricultural practices. Biochar application in the field significantly increased the tomato VC by 20.29% (95% CI: 1.47% to 42.61%) but was not significant in the greenhouse. When the application rate of biochar was at ≤5, 20~50, and >80 t·hm^{-2}, tomato VC was significantly increased by 14.11% (95% CI: 2.79% to 26.67%), 17.00% (95% CI: 3.42% to 32.37%), and 16.00% (95% CI: 6.57% to 26.26%), respectively, while the variation in the other groups was not significant. Regarding experimental duration, biochar application decreased the VC of tomato fruits in all groups but not significantly.

3.4. Meta-Analysis of Soil Properties

According to the collected data, Figure 5a shows the effects of the soil properties on yield. The tomato yield increased by 20.48% to 36.53%, depending on the soil pH. The greatest promotion of yield was observed for a soil pH above 7.5 (36.53%; 95% CI = 30.12%, 43.27%), followed by soil pH 6.5~7.5 and ≤6.5. With regard to SOM, the greatest increase in yield was observed when SOM was below 9.44 g·kg^{-1} (45.20%; 95% CI = 36.53%, 54.42%), and the lowest increase was observed when SOM > 16.80 g·kg^{-1} (22.64%; 95% CI = 13.73%, 32.24%). Regarding EC, the greatest increase in the tomato yield was observed at EC ≤ 142 μS·cm^{-1} (35.32%; 95% CI = 28.33%, 42.68%). Regarding soil available potassium, the greatest increase in yield was observed at ≤101.0 mg·kg^{-1} (43.93%; 95% CI = 33.52%, 55.14%). Regarding soil available phosphorus, the greatest increase in yield was observed at ≤12.6 mg·kg^{-1} (52.77%; 95% CI = 42.28%, 64.04%). Regarding soil total nitrogen, the greatest increase in yield was observed at ≤0.96 g·kg^{-1} (37.81%; 95% CI = 27.85%, 48.67%). Regarding soil bulk, the greatest increase in yield was observed at 1.38~1.41 g·m^{-3} (51.15%; 95% CI = 36.60%, 67.25%).

Figure 5b shows the effects of the soil properties on the sugar/acid of tomato fruits. When soil pH ≤ 6.5, sugar/acid was significantly increased by the application of biochar (18.75%; 95% CI = 3.20%, 36.64%). In contrast, a significant decrease was observed when soil pH > 7.5 (−13.18%; 95% CI = −18.80%, −7.17%). When SOM was between 9.44 and 16.80 g·kg^{-1}, sugar/acid was significantly decreased by the application of biochar (10.30%; 95% CI = 2.06%, 17.85%). Regarding EC, the increase in sugar/acid was significant at >185.0 μS·cm^{-1} (16.24%; 95% CI = 1.86%, 32.64%) but was not significant at 142.0~185.0 μS·cm^{-1} (−4.63%). When the soil available potassium was below 101.0 mg·kg^{-1}, the application of biochar significantly reduced the sugar/acid by 14.62% (95% CI = −26.65%, −1.95%), but was not significant at 101.0~155.6 mg·kg^{-1} (−0.54%) and >155.6 mg·kg^{-1} (6.18%). When the soil available phosphorus was at 12.64~81.26 mg·kg^{-1}, the decrease in sugar/acid was significant (−25.26%, 95% CI = −33.93%, −15.46%), but was not significant at ≤12.64 mg·kg^{-1} (4.28%) and >81.26 mg·kg^{-1} (5.28%). Regarding soil bulk, the decrease

in sugar/acid was significant at 1.38~1.41 g·m^{-3} (51.15%; 95% CI = 36.60%, 67.25%), but was not significant at ≤1.38 g·m^{-3} (−0.50%) and >1.41 g·m^{-3} (3.43%).

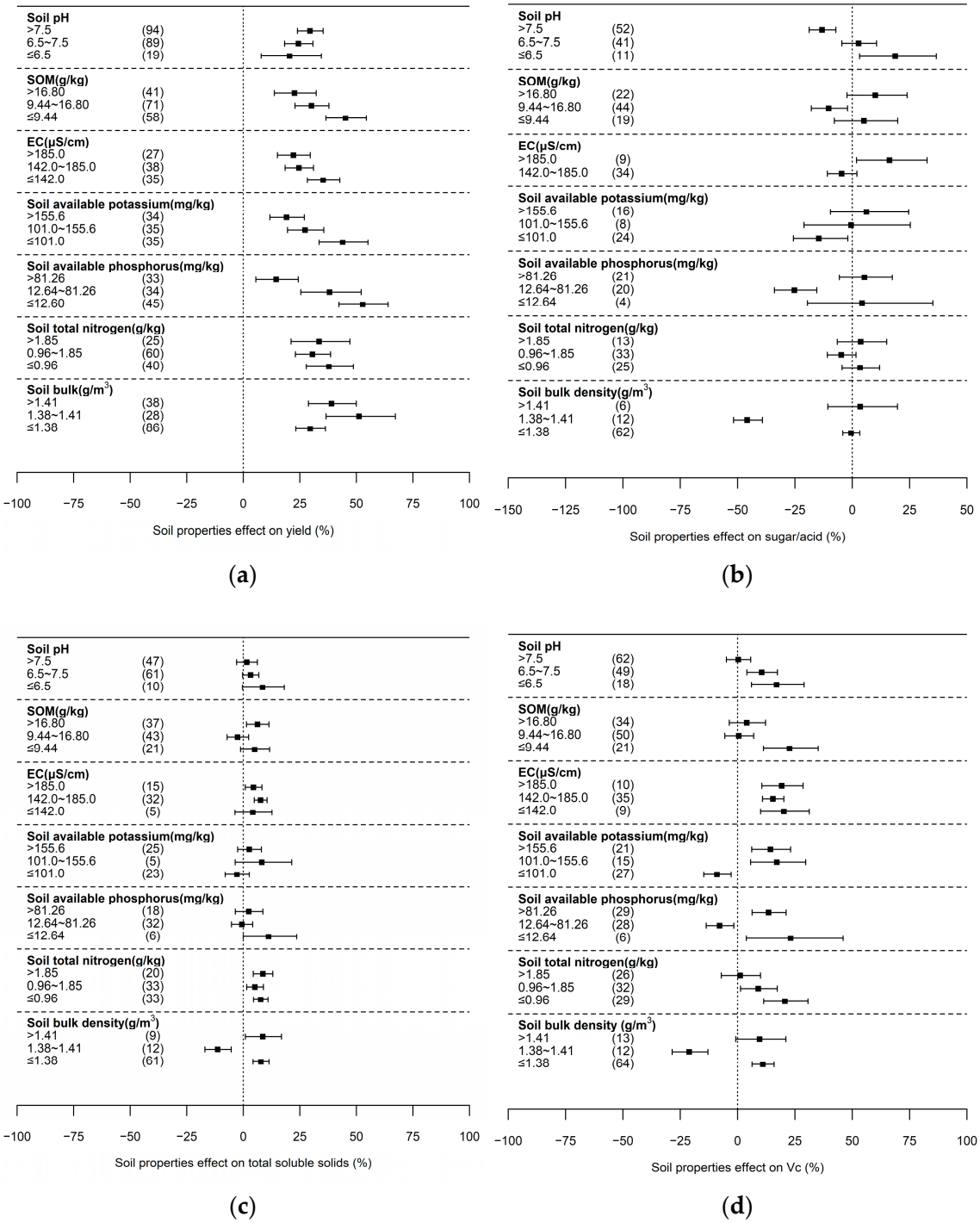


Figure 5. A forest plot shows the percentage changes (%) of biochar effects on (**a**) tomato yield, (**b**) sugar/acid, (**c**) TSS, and (**d**) VC in response to different soil properties. Bars indicate 95% confidence intervals. Data in parenthesis represent the number of observations.

The effects of the soil properties on the TSS of tomato fruits are given in Figure 5c. When SOM > 16.80 g·kg^{-1}, TSS was significantly increased by the application of biochar (6.28%; 95% CI = 1.42%, 11.36%), but was not significant at ≤9.44 g·kg^{-1} (5.01%) and >9.44~16.80 g·kg^{-1} (−2.50%). Regarding EC, the increase in TSS was significant at >185.0 μS·cm^{-1} (4.50%; 95% CI = 0.88%, 8.25%) and 142.0~185.0 μS·cm^{-1} (7.65%;

95% CI = 4.84%, 10.53%), but was not significant at ≤142.0 μS·cm^{-1} (4.19%). When the soil available phosphorus was at ≤12.64 mg·kg^{-1}, the increase in the TSS was significant (11.18%, 95% CI = 0.001%, 23.61%), but was not significant at 12.64~81.26 mg·kg^{-1} (−0.64%) and >81.26 mg·kg^{-1} (2.40%). Tomato TSS increased by 5.15% to 8.65%, depending on the soil total nitrogen. The greatest promotion of TSS was observed at a soil total nitrogen above 1.85 g·kg^{-1} (9.65%; 95% CI = 4.42%, 13.06%), followed by ≤0.96 g·kg^{-1} and 0.96~1.85 g·kg^{-1}. Regarding soil bulk, the increase in TSS was significant at ≤1.38 g·m^{-3} (7.79%; 95% CI = 4.30%, 11.40%) and >1.41 g·m^{-3} (8.61%; 95% CI = 0.93%, 16.89%), but was not significant at 1.38~1.41 g·m^{-3} (−11.33%).

Figure 5d displays the effects of the soil properties on the VC content of tomato fruits. Regarding soil pH, the increase in VC was significant at ≤6.5 (16.96%; 95% CI = 6.06%, 28.99%) and 6.5~7.5 (10.51%; 95% CI = 4.06%, 17.37%), but was not significant at >7.5 (0.28%). When SOM ≤ 9.44 g·kg^{-1}, VC was significantly increased by the application of biochar (22.64%; 95% CI = 11.30%, 35.15%), but was not significant at >16.80 g·kg^{-1} (0.53%) and >9.44~16.80 g·kg^{-1} (3.98%). Tomato VC increased by 15.43% to 20.17%, depending on EC. The greatest increase in VC was observed at EC ≤142.0 μS·cm^{-1} (20.17%; 95% CI = 10.03%, 31.25%), followed by >185.0 μS·cm^{-1} and 142.0~185.0 μS·cm^{-1}. After biochar was applied, except for soil available potassium ≤101.0 mg·kg^{-1}, available potassium 101.0~155.6 and >155.6 mg·kg^{-1} showed significant positive effects on tomato VC, with an increase of 17.07% (95% CI = 5.68%, 29.70%) and 14.32% (95% CI = 6.16%, 23.11%) relative to the control treatment. Also, the effects on VC were significantly increased by soil available phosphorus >81.26 mg·kg^{-1} (13.51%; 95% CI = 6.33%, 21.16%) and ≤12.64 mg·kg^{-1} (23.12%; 95% CI = 3.82%, 46.01%). When soil total nitrogen was below 1.85 g·kg^{-1}, VC was significantly increased by the application of biochar. In the group >1.85 g·kg^{-1}, biochar application increased the tomato VC but not significantly. When soil bulk ≤1.38 g·m^{-3}, VC was significantly increased by the application of biochar (11.01%; 95% CI = 6.31%, 15.92%). In contrast, a significant decrease was observed when soil bulk was 1.38~1.41 g·m^{-3} (−21.11%; 95% CI = −28.52%, −12.93%).

4. Discussion

4.1. Effect of Biochar Properties

The inherent properties of biochar are the fundamental reasons for its impact on crop yields [43]. Although biochar has undergone the processes of crushing and pyrolysis, it still retains the basic appearance of the raw materials in terms of physical structure, and it also inherits the elemental composition characteristics of the raw materials in terms of chemical composition [44]. Its physical structure and chemical composition determine the potential of biochar to affect soil ecosystems [45]. This study shows that, apart from sludge biochar, biochar from wood, straw, shell residues, and mixed materials all demonstrate a positive effect on tomato yield enhancement. Simultaneously, biochar produced by wood and straw can significantly improve the TSS and VC in fruits. Wood and straw biochar, with its porous nature and large surface area, can absorb various substances including minerals and organic compounds with diverse molecular dimensions and chemical properties [45]. The biochar derived from wood and straw exhibited a comparatively elevated carbon-to-nitrogen (C/N) ratio compared to other types [46], potentially creating an advantageous environment for the development of fungal proliferation [44]. Additionally, wood biochar contained a significantly higher amount of ferric ions than the other biochar [46], enhancing SOM adsorption onto clay minerals and improving soil structure [47]. Therefore, the type of biochar feedstock has played a significant role in the variation in outcomes, with biochar produced by wood and straw leading to the greatest enhancement in yield and quality of tomatoes compared to all other biochar when we consider the utilization of biochar in tomato cultivation.

In addition to the biochar feedstock types, the pyrolysis temperature is another crucial determinant of biochar's physicochemical characteristics. The meta-analysis conducted reveals that biochar produced at pyrolysis temperatures below 400 °C can, on average,

increase tomato yields by 40.97%, and as the pyrolysis temperature rises, the effects of biochar on promoting crop yield gradually diminishes, while biochar with pyrolysis temperatures between 401 and 500 °C exhibits a more pronounced improvement in the quality indicators of TSS and VC than biochar produced at 400 °C. The results were consistent with the research by Suthar et al. [25], which investigated the impact of bamboo biochar pyrolyzed at different temperatures on the growth and fruit quality of tomato, for that soil applied with biochar at a lower pyrolysis temperature contained higher contents of NO_3-N, P, Ca, and Mg, thereby proving more advantageous for tomato cultivation. The biochar's spacial structure and composition change with varying pyrolysis temperatures, and as indicated by prior research, the biochar's pore structure generates at a high pyrolysis temperature [48]. However, once the temperature surpasses a certain threshold, the tubular structures within the biochar can buckle, creating a layered arrangement that leads to the obstruction of micropores and a consequent decrease in nutrient absorption capacity, and this structural change in biochar may limit its ability to retain and release nutrients, thereby having an adverse effect on crop growth [49]. Throughout the pyrolyzation, nitrogen content progressively decreases with the increase in pyrolysis temperature; therefore, biochar produced through high-temperature pyrolysis lacks nitrogen and cannot effectively replenish nutrients in the soil, particularly in infertile soils. Additionally, biochar contains biodegradable organic carbon and ash, which can activate the decomposition of soil microorganisms, provide nutrients, and effectively improve crop yield and quality. Some studies have shown that biochar pyrolyzed at temperatures > 500 °C almost lack labile organic carbon, while those pyrolyzed at temperatures below 500 °C contain labile organic carbon [50,51]. The conducted meta-analysis suggests that biochar generated within the temperature range of 401~500 °C tends to be the most effective in improving tomato yield and fruit quality.

4.2. Effect of Biochar pH and Soil pH

The pH of biochar and soil are pivotal factors influencing soil condition and, consequently, plant growth via affecting the microbial community and nutrient cycle [52]. Most biochar varieties exhibit alkaline properties, which contributes to the liming effect [53], considered one of the key mechanisms for enhancing plant productivity [54]. However, the application of biochar to alkaline soils may lead to an excessive increase in soil pH, potentially inhibiting plant growth due to nutrient precipitation and reduced availability. This meta-analysis indicate that the effects of biochar on sugar/acid, TSS, and VC are more significant in acidic soils compared to neutral and alkaline soils, consistent with previous and present studies [12,25,55]. However, yields are found to be higher in alkaline soils compared to both neutral and acidic soils, not entirely aligning with previous research [8,56]. The reason for the discrepancy is because the data collected on alkaline soils primarily pertain to experiments on tomato cultivation in saline–alkali land or with irrigation water that has a high salt content. Excessive salt can lead to salt accumulation in tomato plants, thereby restricting their growth. Biochar could adsorb salts from the soil, which result in a higher improvement of tomato yields under alkaline soil conditions [19]. Furthermore, according to Dai et al., the improvements in alkaline soil characteristics such as SOM, EC, C/N, and cation exchange capacity (CEC), through the application of biochar, are likely the primary factors responsible for the positive effects on yield, more so than the alterations in soil pH. Additionally, it was also observed that the tomato yield in the application of biochar with high pH (>10) was higher than that with low pH (8.5~10 and $\leq$8.5). In contrast, the quality indicators, TSS, sugar/acid, and VC, exhibit an opposite trend. This could be due to the liming effect of biochar, which increases tomato yield, but applying biochar with a high pH may influence the activity of soil microbes, reducing the utilization of nutrients, and consequently leading to a decline in tomato quality. Although biochar may play a positive role in increasing tomato yields under alkaline soil conditions, its elevation of soil pH could be detrimental to the maintenance of crop quality. To achieve the sustainable use of biochar and mitigate its potential negative impacts on soil pH, a careful

assessment of biochar properties is required, along with consideration of its application method and dosage. For instance, adjusting the amount of biochar applied or selecting biochar suitable for specific soil types could optimize soil pH, thereby enhancing both crop yield and quality.

4.3. Effect of Agriculture Practices

An increment in tomato yield was observed with increasing rates of biochar application. Additionally, the higher application rates of biochar significantly improved the TSS content in tomato fruits. Earlier research has indicated that a higher biochar application rate improved organic matter, nitrogen, phosphorus, and other essential nutrients in the soil [57,58], which accordingly increased plant yield and quality. Zhang et al. [59] examined the impact of varying rates of biochar (0, 22.5, 67.5, and 112.5 t/ ha) on tomato yield, and observed that yield increased along with the rate of biochar application. Nonetheless, it is imperative to weigh the economic advantages, specifically the input–output ratio, and to choose the biochar application rate accordingly.

VC is one of most important antioxidants and represents the main nutritional quality of tomato [60]. According to this meta-analysis, the increase in VC was more significant in field-grown tomatoes than greenhouse-grown tomatoes. The heterogeneity observed between field-grown and greenhouse-grown tomatoes can be attributed to disparities in planting times, which can alter the intensity, duration, and quality of light and temperature exposure. These factors may consequently impact the levels of antioxidants present in the tomato fruits [61].

We found that the positive effects of biochar application on tomato yield, TSS, and sugar–acid ratio decreased as the duration of the experiment extended. This could be due to the biochar's liming effect diminishing over time, as well as a reduction in biochar's adsorption potential for ions due to oxidation [62]. The aging process of biochar in the soil results in alterations to its surface properties through oxidation, the formation of oxygen-containing functional groups, and the adsorption of natural organic substances [63]. These changes can cause the pores of the biochar to become blocked and lead to the development of organic mineral complexes that cover the biochar's surface [64]. Therefore, for long-term tomato cultivation, it is recommended to periodically replenish biochar to ensure the sustainability of its effects. Despite these benefits, the application of biochar must also consider the potential negative environmental impacts it may bring, especially increased soil erosion. The application method and particle size of biochar can significantly affect soil erosion rates [65]. Fine biochar particles are more susceptible to wind erosion, while larger particles can be removed by water, leading to a loss of topsoil and associated nutrients [66].

4.4. Effect of Soil Properties

Nitrogen plays an indispensable role in the growth of tomato, for it is the essential element and necessary component for tomato. The frequent reports of improved plant yield following biochar application are attributed to enhanced soil nitrogen retention and utilization efficiency, which are a result of the additional nitrogen supply and the biochar's positive impact on the nitrogen cycle [67–69]. Meanwhile, the high adsorption capacity of biochar may lead to the fixation of nutrients, thereby reducing the nutrient availability for plants and microorganisms. Although the application of biochar can increase the total nitrogen content in the soil, the bioavailable nitrogen does not significantly increase, and this may even lead to nitrogen immobilization in the soil, diminishing the nitrogen availability for plants [70]. In the present study, as soil nitrogen levels become more deficient, the effects of biochar application on increasing tomato yield and VC content become increasingly significant. Biochar might influence the soil nitrogen cycle through various pathways such as (1) modifying the soil's inorganic nitrogen content through biochar's nitrogen adsorption or desorption [71], (2) affecting the microbial activities of nitrogen mineralization or immobilization by altering the availability of soil mineralizable

substrates [72], and (3) shifting the equilibrium between nitrification and denitrification processes by changing soil properties like pH and aeration [73].

Biochar is considered a vital source of phosphorus (P) and potassium (K) for plant growth in soils that are deficient in these nutrients, for the application of biochar can significantly improve the contents of soil available P and K [74]. The meta-analysis results showed that the biochar applied in soil with low P and K has a more significant improvement on the yield and VC of tomato than those applied in soil with high P and K. The research of Zheng et al. [44] has indicated that in soils amended with biochar, there is an increased abundance of phosphorus-solubilizing bacteria, such as Pseudomonas and Bacillus. Consequently, the fixed forms of phosphorus present in soil minerals, SOM, or biochar can be dissolved or converted into bioavailable phosphorus. Thus, employing biochar rich in phosphorus and potassium could serve a dual function as both a nutrient provider and improver.

SOM plays a crucial role in supporting plant growth and maintaining soil fertility through several key functions such as nutrient supply, water retention, and an energy source for microorganisms [75,76]. In this study, we found that biochar applied to soil with low SOM leads to a more notable increase in tomato yield and VC levels than when it is applied to soil with a high SOM content. Biochar can influence the decomposition of SOM through a priming effect, which could enhance soil fertility and crop yields by stimulating microbial activity and accelerating decomposition, thereby making nutrients more available to plants [77].

Typically, biochar's low bulk density and well-developed porous structure could potentially enhance the structural characteristics of amended soils, including soil bulk density, porosity, pore size distribution, and aggregate stability [78,79]. Additionally, the structural enhancement can further boost the soil's water-holding capacity, enhance the nutrient cycle, and activate microorganisms, all of which are conducive to plant growth [80]. In this study, the positive effects of biochar application are more significant on tomato yield in high-level bulk density (>1.41 $g \cdot m^{-3}$) and medial-level bulk density (1.38~1.41 $g \cdot m^{-3}$) soils compared to low-level bulk density ($\leq$1.38 $g \cdot m^{-3}$) soil. It indicates that the application of biochar has a certain amendatory effect on soils with a higher bulk density. By reducing bulk density, it can improve soil structure and enhance tomato production.

The incorporation of biochar into soil can significantly alter the structure and function of soil microbial communities, not only by amending the soil's physical and chemical properties but also by potentially changing the levels of toxic substances present in the soil. Biochar may release or adsorb pollutants such as polycyclic aromatic hydrocarbons (PAHs), volatile organic compounds (VOCs), dioxins, and potentially toxic elements (PTEs) during its production process [81]. The presence of these contaminants could exert toxic effects on soil microorganisms, affecting their metabolic activities and biogeochemical cycling processes [82]. Specifically, the presence of PAHs in biochar, known for their mutagenic and carcinogenic properties, may negatively impact soil microbial communities [83]. The accumulation of PAHs could disrupt the metabolic pathways of soil microorganisms, reducing their ability to decompose organic matter and transform nutrients. Similarly, VOCs present in biochar might inhibit the enzymatic activities of soil microorganisms and alter nitrogen cycling processes [84]. Furthermore, the presence of dioxins and PTEs in biochar could change the composition and diversity of soil microbial communities by interfering with their physiological functions [85].

4.5. Limitations and Prospects

In this meta-analysis on the impact of biochar application on tomato yield and fruit quality, we recognize several limitations. The stringent criteria for the study selection resulted in a reduction from an initial pool of numerous articles to a final set of 46 articles included in the meta-analysis (Supplementary Material). This curation, while necessary for methodological rigor, potentially excludes valuable data and perspectives that could enrich our findings. While the complexity of field conditions and the intricate interactions

between biochar and soil properties are challenging to fully capture through meta-analysis, the variability in biochar production methods, application rates, and soil types across the included studies means that our results may not be universally applicable. Moreover, the specific conditions such as soil pH and organic matter content, which are critical for understanding biochar's effects, were not uniformly reported across studies, leading to potential gaps in our analysis. Furthermore, our meta-analysis primarily reflects short-term outcomes, with a preponderance of studies reporting effects within the first year after biochar application. This focus on shorter duration means that our understanding of long-term effects, including changes in soil fertility and structural properties over time, is limited. The long-term aging process of biochar in the soil and its implications for sustainable agricultural practices require further investigation. Lastly, economic considerations are paramount for the practical adoption of biochar in agriculture. The financial implications of biochar application at various rates and its return on investment in terms of yield and quality improvements were beyond the scope of our meta-analysis but are crucial for making biochar a viable option for farmers.

To overcome these limitations, future research should include long-term field studies to assess the enduring impact of biochar on soil properties and tomato yield. Conducting on-site experiments will be crucial for validating the findings presented in this meta-analysis, ensuring that the results are applicable in real-world agricultural settings. Moreover, studies that explore the integration of biochar with other agricultural practices are needed to provide a comprehensive understanding of its role in sustainable agriculture and to guide its practical application.

5. Conclusions

In this study, we investigated the effects of biochar application on tomato yield and fruit quality through a meta-analysis. As a result, we found that the application of biochar can substantially improve tomato yield and the nutritional attributes of the fruit. Specifically, the application of biochar resulted in a significant increase in tomato yield by 29.55% and improved the content of TSS by 4.28% and VC by 6.77%, with higher application rates showing more significant effects. However, the benefits may diminish over time due to the aging of biochar in the soil, necessitating periodic replenishment to maintain its positive effects. The analysis of biochar properties showed that the biochar type and pyrolysis temperature significantly influenced the outcomes, with wood and straw biochar and those pyrolyzed at temperatures between 401 and 500 °C being the most effective for elevating tomato yield and quality. The study also indicated that the soil's initial properties, particularly SOM, pH, and nutrient levels, interacted with biochar to affect yield and quality. It was observed that biochar application was particularly beneficial for soils with a high bulk density, low SOM, and nutrient deficiency. In conclusion, biochar can serve as a multifunctional amendment in tomato agriculture; the properties of biochar should be selected carefully prior to application according to the conditions of targeted soils and agricultural practices.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/su16156397/s1.

Author Contributions: Data collection and collation, Y.Y., S.S. and M.W.; Responsible for manuscript writing and chart making, Y.L. and L.X.; Responsible for language editing and text modification, C.X. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Key Research and Development Projects in Shanxi Province, grant number 202102140601015; the Shanxi Provincial Natural Science Youth Fund, grant number 20210302124152; and the Fund Program for the Scientific Activities of Selected Returned Overseas Professionals in Shanxi Province, grant number 20240058.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available upon request.

Conflicts of Interest: The authors declare no conflicts of interest.

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